

NANDA KUMAR

Machine Learning Engineer | Generative AI & Computer Vision | Diffusion Models · Production VFX-AI
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PROFESSIONAL SUMMARY

Machine learning engineer specializing in generative AI and computer vision, with a Computer Science foundation and a continuous technical thread across 12+ years — from large-scale DevOps at Zoho (SaaS at 10M+ users) to ~2 years building production Generative AI (Stable Diffusion, ControlNet, custom computer vision) for live feature-film work. A parallel career as a feature-film Director of Photography runs alongside the engineering, adding practitioner-level color science and visual-quality judgment that few engineers can bring to diffusion and computer-vision pipelines.

CORE SKILLS

Diffusion & Generative Media: Stable Diffusion (1.x / XL), ControlNet, ComfyUI, LoRA / fine-tuning, model quantization, TensorRT

Computer Vision: SAM2 segmentation, rotoscoping, optical flow, temporal-consistency methods, OpenCV, object detection

Machine Learning / Deep Learning: PyTorch, Hugging Face (Transformers, Diffusers), model training & deployment, GPU/CUDA inference

MLOps & Infrastructure: Docker, FastAPI, REST APIs, checkpoint versioning, Linux, Git, production release management

Additional Skills

LLM / VLM: Local VLM (Ollama — LLaVA / Qwen-VL) auto-annotation, LangChain orchestration

Image / Video model tooling: Qwen-Image, LTX, Seedance, Higgsfield, Luma Labs (evaluated in production experiments)

Languages & Tools: Python, JavaScript, Bash, SQL, NumPy, FFmpeg

PROFESSIONAL EXPERIENCE

Founder & CTO — Neural Engine Technology Private Limited

Feb 2025 – Present

Chennai, India | AI-VFX venture — shipped the entire GenAI pipeline (founding team), research to production

- Shipped a diffusion pipeline (Stable Diffusion + ControlNet + ComfyUI) into live feature-film post-production.
- Replaced multi-hour manual rotoscoping with a model-driven SAM2 segmentation + temporal-consistency module, cutting manual roto time by ~75% per shot while producing film-grade output.
- Designing AI-VFX systems and post-production workflows for feature films.
- Led the full AI product lifecycle — technical vision, founding-team hiring, ML research to deployment, vendor selection, and stakeholder communication.

Tech: Python, PyTorch, Stable Diffusion, ControlNet, ComfyUI, LoRA / fine-tuning, model quantization, SAM2, OpenCV, CUDA/GPU, FastAPI, Docker, REST APIs

Director of Photography — sixtyninemm Films

Feb 2017 – Feb 2025

Chennai, India | Feature & short films

- Cinematographer on feature and short films, including the 2024 Tamil theatrical feature *Bayamariya Brammai* (dir. Rahul Kabali, 69 MM Films); owned the full visual language — lensing, lighting, framing, on-set color management, and the final color correction (grade) of the film — from pre-production through delivery.
- Worked closely with directors and post teams across the production lifecycle, driving look development, on-set creative decisions, and color continuity into the grade and final output.
- Practitioner-level color science, image-quality judgment, and shot intuition are now applied directly to AI-VFX system design and visual-quality evaluation — grounding the media pipelines I build in how footage is actually shot and finished.

Member of Technical Staff, Service Delivery (DevOps) — Zoho Corporation

Nov 2014 – Feb 2017

Chennai, India | Enterprise SaaS

- Owned production deployments and release rollouts for one of India's largest SaaS suites (10M+ users), delivering reliable, low-downtime releases across multiple product surfaces.
- Handled server provisioning, infrastructure monitoring, and grid operations to keep large-scale production services healthy and available.

KEY AI PROJECTS

- UNIRAW — Universal RAW Compression & Intelligence Engine** (Founder / Lead Engineer — github.com/smnanda23/uniraw). Privacy-first RAW archival engine compressing Nikon NEF / DNG masters ~20-50x (about 12 MB to 0.7 MB) while reconstructing editable, chemically accurate DNGs read as native RAW by Lightroom/Photoshop. Pipeline: float Lightmap normalization + K-means 4x4 vector quantization + residual encoding + LZMA packaging, with fully offline VLM auto-annotation (Ollama — LLaVA / Qwen-VL). React/Vite + Flask/FastAPI stack.
- AI-VFX Production Pipeline:** End-to-end diffusion pipeline (SD + ControlNet + ComfyUI) for feature-film post; custom ControlNet conditioning preserves shot geometry across frames, replacing hours of manual cleanup per shot.
- Video Segmentation & Intelligent Rotoscoping:** CV module combining SAM2-based segmentation with temporal-consistency smoothing for film-grade output, eliminating manual rotoscoping from the pipeline.
- Generative Look-Development Tool:** Stable Diffusion–based concept-art generator letting directors/DPs iterate on visual references in minutes rather than days.

EDUCATION & CERTIFICATIONS

B.E. Computer Science — Sri Muthukumaran Institute of Technology, Chennai (2012)

Certifications: DeepLearning.AI — Generative AI Specialization; OUTSKILL — Generative AI Mastermind & AI Engineering Mastermind